enough red-fuming nitric acid, which was then used as the liquid propellant, he bent over the storage tank. As the name suggests, the acid when exposed to air, gives out noxious red fumes. Pillai fainted and had to be taken to a hospital for first-aid. All hell broke loose. The see-we-told-you chorus became unbearable.

What they probably knew was that such propellants as red fuming nitric acid are kept in large barrels even on college campuses. In fact, testing a liquid engine is safer than a solid test: a propellant flow can be stopped at will, if something goes wrong in a liquid engine. This is not possible with a solid engine, which is like igniting compressed gunpowder. I would not give up, and decided to test the engine.

They suggested that I take the engine to Sriharikota for further tests. It was not a bad idea, since Sriharikota had much more uninhabited space. I agreed, though I realised that their idea was to drive the liquid engine out of Thumba, than to offer a better alternative. The island with Bay of Bengal on a side and a lagoon on the other three sides, was indeed a safer place, but Sriharikota had no facilities for our stay: There were a couple of rooms for our team of more than twenty people.

The first test was a resounding failure, literally. An explosion ripped through the engine, smashing the test stand that holds the engine to smithereens. I was devastated.

A failure analysis later showed that the reason was the water we used to calibrate the system before filling fuel. To save fuel, water is passed from a tank to the engine, with several filters on the way, to gauge the flow of the liquid. Through mathematical calculations, we would make corrections based on the difference in the viscosity and other parameters of water, before cleaning and filling the system with the liquid fuel in place of water. Our calculations were based on the presumption that the water is pure. It was not. Fine suspended particles in water settled as sediments on the filters and the readings we got were erroneous. When we filled the fuel and fired the engine, we were unaware that the anomalies in the calculations would result in a heavier flow of the liquid fuel. The result was the disaster.